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A Polyimide Based Force Sensor Fabricated Using Additive Screen-Printing Process for Flexible Electronics

DINESH MADDIPATLA[1,*], (Student Member, IEEE), XINGZHE ZHANG[1,*], ARNESH K. BOSE[1], (Graduate Student Member, IEEE), SIMIN MASIHI[1], (Graduate Student Member, IEEE), BINU B. NARAKATHU[1], (Member, IEEE), BRADLEY J. BAZUIN[1], (Member, IEEE), JOHN D. WILLIAMS[2], MICHAEL F. MITCHELL[2], AND MASSOOD Z. ATASHBAR[1], (Senior Member, IEEE)

[1]Department of Electrical and Computer Engineering, Western Michigan University, Kalamazoo, MI 49006, USA
[2]Additive Electronics Technology (AET) Group, Boeing Research and Technology, The Boeing Company, Huntsville, AL 35824, USA

Corresponding author: Dinesh Maddipatla (dinesh.maddipatla@ieee.org)

*Dinesh Maddipatla and Xingzhe Zhang contributed equally to this work.

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ABSTRACT A flexible capacitive force sensor was developed for force sensing applications. The force sensor consists of two electrodes and a dielectric layer. The electrodes were fabricated by depositing conductive silver (Ag) ink on a flexible polyimide platform using screen-printing process. The dielectric layer was prepared by mixing the contents of polydimethylsiloxane (PDMS) (pre-polymer and curing agent) in a 16:1 ratio. Then the PDMS dielectric layer was sandwiched between the screen-printed Ag electrodes. The capability of the fabricated force sensor was investigated by recording its capacitive response for varying applied forces of 100 N. It was observed that the capacitance increased from 5.83 pF to 6.36 pF as the force was increased from 0 N (no load) to 100 N. A sensitivity and correlation coefficient of 0.081%N^{-1} and 0.998 were calculated for the force sensor. In addition, a response time and recovery time of 3.7 seconds and 5.7 seconds was measured for the fabricated force sensor. The relative humidity (% RH) tests performed from 20% RH to 80% RH, in steps of 20% RH, revealed that there was a minimal effect of RH on the base capacitance of the force sensor at room temperature.

INDEX TERMS Additive screen-printing process, capacitive based pressure sensor, polyimide substrate, PDMS dielectric layer, flexible.

I. INTRODUCTION

In recent years, the need for the development of flexible force sensors has been increased due to its wide range of application in infrastructure monitoring, robotics, electronic skins, health care and diagnostic purposes, and smart displays [1]–[5]. Piezoelectricity, piezo resistivity, and capacitance are the three commonly employed force sensing mechanisms. Among them, capacitance-based sensing mechanisms are widely preferred due to its multiple advantages including relatively less power consumption, repeatability of the responses, less prone to humidity and temperature changes, and less susceptibility to noise [6]–[9]. The force sensors are typically fabricated using traditional/conventional methods that often employs rigid platforms with either hanging or cavity structured designs [10]–[15]. Thus, these sensors do not offer the required flexibility and conformability for the emerging smart wearable industry, specifically e-skins [10]–[15]. In addition, these force sensors are not suitable for asset/structural health monitoring of aircrafts or any civil structures due to the limitations associated with conformability as well as less resistance towards harsh environments including variations in humidity levels. These drawbacks can be alleviated by exploring some novel fabrication techniques such as additive print manufacturing processes that can enable the development

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of relatively cost-effective flexible and conformal force sensors.

Additive print manufacturing processes such as screen, inkjet, flexography and gravure provides numerous advantages for the fabrication of various electronic devices including device flexibility, conformability, low wastage of materials as well as processing temperatures, relatively fewer and simpler fabrication steps [16]–[31]. In addition, these processes have roll-to-roll (R2R) capability that enables the large-scale manufacturing of functional devices in a relatively short time and this reduces the production cost [32]–[34]. Among the additive manufacturing process, screen printing process is the most reliable and durable deposition method with high yields over longer print runs providing relatively thicker ink layers [32].

The working principle/mechanism of a capacitive force sensor is same as a parallel plate capacitor and can be mathematically expressed using Eq. (1),

$$C = \varepsilon_r \varepsilon_0 \frac{A}{d} \tag{1}$$

where C - capacitance of the force sensor, ε_r- dielectric constant of dielectric layer, ε_0- dielectric constant of vacuum, A - overlapping electrodes area, d - distance between the electrodes (thickness of the dielectric layer). When an external force is perpendicularly applied on the force sensor, the dielectric layer gets compressed (the thickness of the dielectric layer decreases) and this results in a proportional capacitance change.

There are many studies performed on the development of flexible force sensors using dielectric layers made of polydimethylsiloxane (PDMS) and liquid metals, but the sensitivity of these sensors are relatively low [35]–[37]. Typically, the PDMS dielectric layers employed in these sensors use a standard fabrication method in which the pre-polymer and curing agent of the silicone elastomer kit are mixed in a 10:1 (w/w) ratio. However, if the hardness of the PDMS can be decreased, this can increase the compressibility of the PDMS and results in higher relative capacitance change and increases the sensitivity of the force sensor. The hardness of the PDMS can be decreased by increasing the ratio of pre-polymer content to curing agent during the fabrication process. A PDMS dielectric layer prepared by mixing the contents of silicone elastomer kit, pre-polymer and curing agent in a 16:1 (w/w), will provide high stress to strain ratio and this will result in obtaining higher relative capacitance change [38].

Recently, force/pressure sensors have been fabricated using plastic materials (polyethylene terephthalate (PET)), conductive materials (graphene, copper, carbon nanotubes) and dielectric materials (PDMS) with pores or micro-structures using 3D templates and laser machining [39]–[44]. Even though the sensitivity of some of these sensors is typically high, the structures in the dielectric layers are prone to collapse or break easily and are not suitable for high force measurements. In addition, the fabrication of these sensors is complex and difficult to obtain the uniformity in pores or microstructures resulting in low fabrication yields and inconsistent output performances from the sensor to sensor. The asset/structural health monitoring of aircrafts or any civil structures requires force sensors that are not only conformable and resistant to environmental changes (variations in humidity levels), but also be able to detect high forces while being robust, stable and compatible for mass production with high fabrication repeatability.

In this work, a flexible capacitive force sensor was developed by using a PDMS based dielectric layer prepared by mixing pre-polymer and curing agent in a 16:1 ratio. The top and bottom electrodes were fabricated by depositing silver (Ag) ink on a polyimide substrate using screen printing process. The flexible polyimide substrate was used as substrate for the electrodes since it provides relatively higher thermal stability, tensile strength, chemical resistance as well as excellent mechanical properties when compared to the mostly used other plastic based materials such as PET (force sensor fabricated on PET was demonstrated in our previous work) [45], [46]. The capability of the fabricated highly robust force sensor to detect varying applied forces was demonstrated by investigating its capacitive response. In addition, response and recovery time as well as the effect of varying relative humidity (% RH) and temperature levels on the sensor was investigated and is presented.

II. EXPERIMENTAL

A. CHEMICALS AND MATERIALS

Ag conductive ink (Ag 800) from Kayaku Advanced Materials, Inc. was used for fabricating the electrodes. A 127 μm thick polyimide film, Kapton® 500HN from Dupont™was used as the substrate. Computer numerical control (CNC) machined mold made of aluminium, and Sylgard® 184 PDMS silicone elastomer kit from Dow Corning was used in fabricating the dielectric layer. A 100 μm thick double-sided polyimide tape consisting of 25 μm thick polyimide film and 38 μm of silicone adhesive coating on each side of polyimide film from BERTECH® was used to attach the top and bottom electrodes to the PDMS dielectric layer. A flat flexible connector (FFC) with crimp-based contact termination, from TE Connectivity AMP Connectors (1-88997-2) and male to female jumper wires (PRT-12794) from SparkFun Electronics were used to connect the sensor to an Agilent E4980A precision LCR meter.

B. SENSOR FABRICATION

Figure 1(a) and (b) shows the sensor schematic and its layer assembly, respectively. The design of top and bottom electrode consists of a circle of 15 mm diameter with a contact terminal of length 15 mm. The conductive Ag ink was deposited on the polyimide substrate using a AMI MSP 485 semi-automatic screen printer from Affiliated Manufacturers Inc and a screen with stainless steel mesh (mesh count - 325, wire diameter - 28 μm and deflection angle - 22.5°), fabricated at Microscreen®, LLC. Then the screen-printed Ag ink was annealed at 130 °C for 8 minutes in a VWR

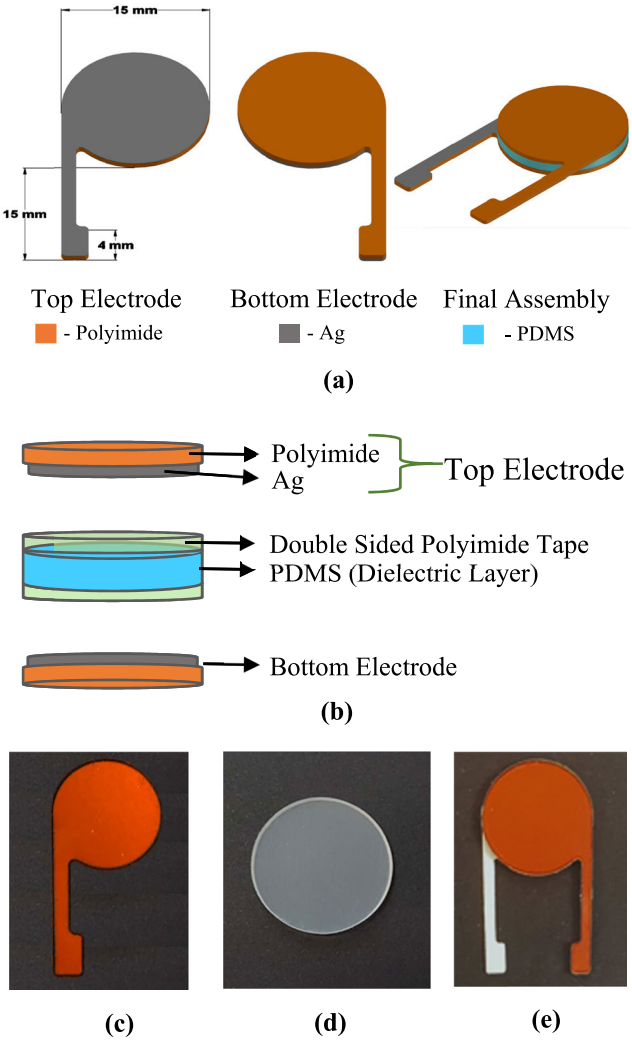

FIGURE 1. **(a) Schematic of the force sensor (Not to scale), (b) force sensor layer assembly, (c) printed electrode, (d) fabricated PDMS and (e) assembled force sensor.**

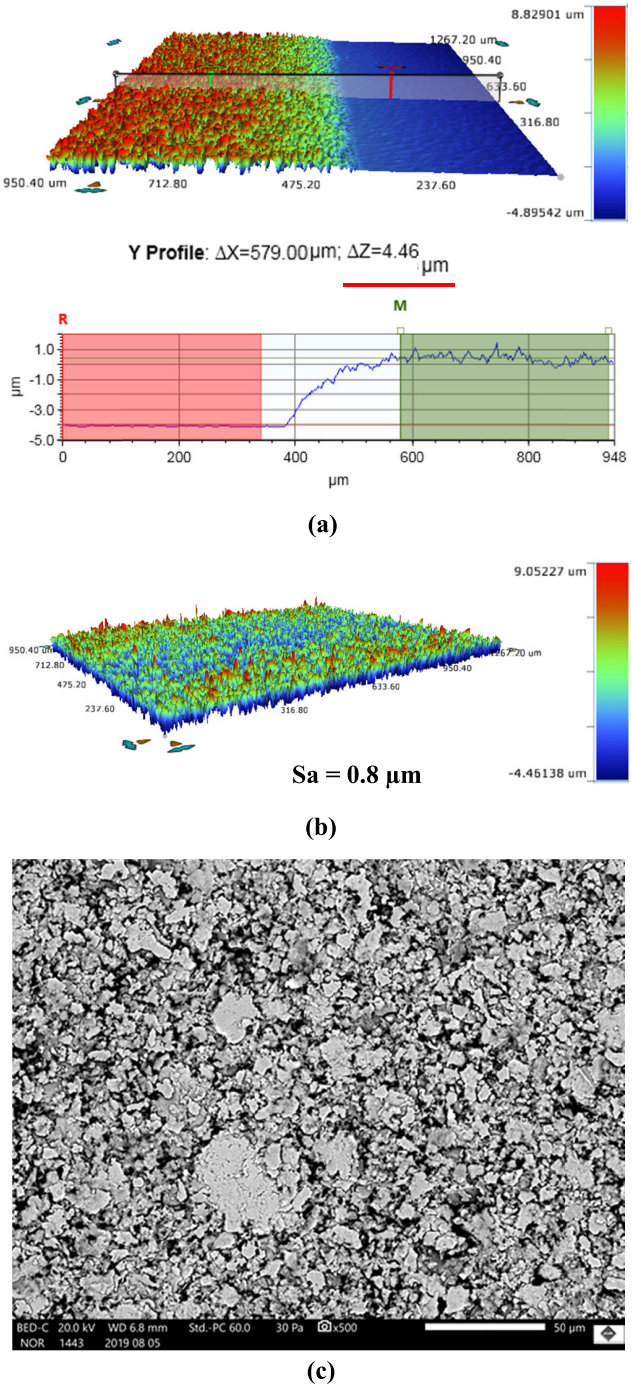

FIGURE 2. **3D output of vertical scanning interferometry of printed Ag on polyimide illustrating an (a) thickness (ΔZ) of 4.4 μm, (b) roughness of 0.8 μm and (c) SEM micrograph of the printed Ag on polyimide substrate.**

1320 thermal oven. The PDMS was prepared by adding the Sylgard® 184 pre-polymer to a curing agent (in a 16:1 (w/w)) and stirring in Thinky AR-100 planetary centrifugal mixer for 3 minutes at 1500 rpm. This resulted in a homogenous mixed PDMS without any bubbles. Following this, the bubble-free PDMS was poured into the 15 mm diameter aluminium mold and annealed for 10 minutes at 110 °C in the VWR 1320 thermal oven. Then, the electrodes (top and bottom) (Fig. 1(c)) and the PDMS dielectric layer (thickness 0.6 mm) (Fig. 1(d)) of the force sensor were then patterned by laser cutting (Universal Laser Systems – PLS6MW) the screen printed samples and PDMS. Finally, the PDMS based dielectric layer with a dielectric constant of 2.65 was sandwiched between the Ag based top and bottom electrodes, using a double-sided polyimide tape to form the robust force sensor as shown in Fig. 1(e) [38].

A thickness of 4.46 μm and roughness of 0.8 μm was measured for the screen-printed Ag layer on polyimide substrate using a Bruker Contour GT-K white light interferometer (5X lens). The 3D output of the white light vertical scanning interferometry with thickness and roughness measurements are shown in Fig. 2(a) and Fig. 2(b), respectively. The print quality of the Ag electrode was examined using a JEOL JSM-IT500HR InTouchScope™ scanning electron microscope (SEM). The SEM micrograph of the Ag electrode was captured in a low vacuum mode (30 Pa)

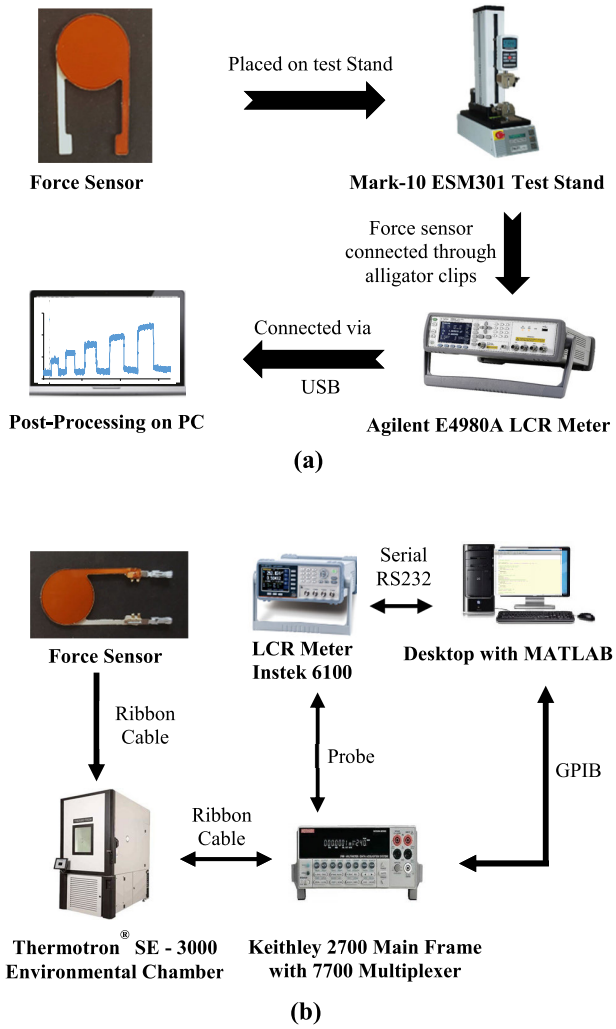

FIGURE 3. Experiment setup to characterize the sensor towards (a) varying applied forces and (b) varying % RH and temperature levels.

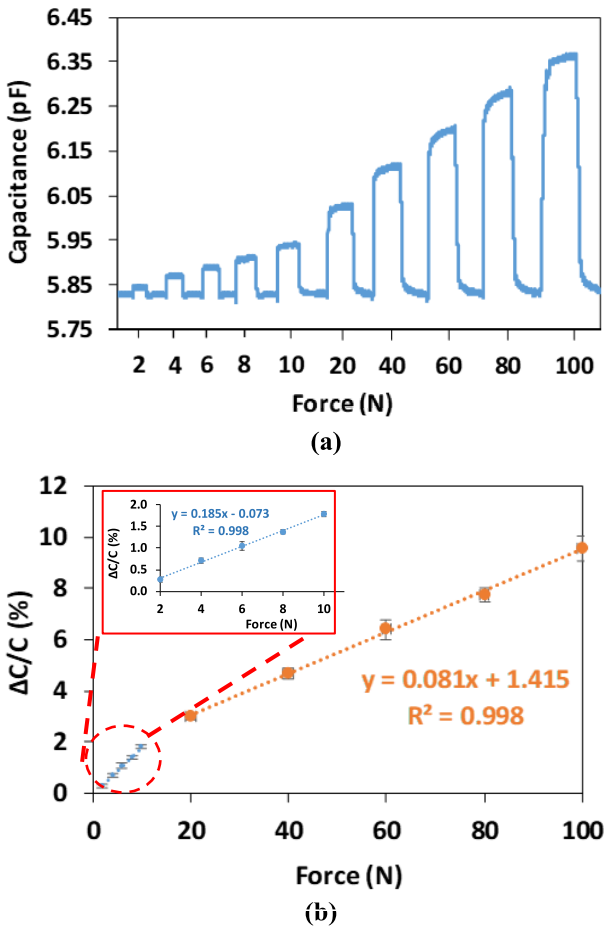

FIGURE 4. (a) Capacitive response of force sensor towards varying forces (up to 100 N) and b) Relative capacitance change of the fabricated force sensor (up to 100 N).

using an accelerating voltage of 20 keV, emission current of 90 μA and backscattered electron detector at a working distance of 6.8 mm. As shown in Fig. 2(c), it is evident that the Ag flakes are homogenously distributed on the polyimide film and connected to each other without any cracks or voids that influences the electrode conductivity.

C. EXPERIMENT SETUP

The experiment setup for characterizing the sensor is shown in Fig. 3(a). The fabricated capacitive force sensor was placed between the clamp holders of a Mark-10 model M5-200 force gauge and Mark-10 ESM 301 motorized test stand. Varying forces up to 100 N were applied perpendicularly to the force sensor using automated Mark-10 force gauge. Then, the sensor with contact crimp socket connectors was connected to an Agilent E4980A precision LCR meter using alligator clips for measuring the capacitive responses of the sensor. An operating voltage and a frequency of 1 V and 1 kHz was applied to the force sensor for capacitive measurements. The LCR meter was connected to a PC with a custom-built LabVIEW™program via a USB cable to automatically record the real-time capacitive responses of the force sensor. Similarly, for measuring the effect of humidity on the force sensors, a Thermotron® SE 3000 environmental chamber was used as shown in Fig. 3(b). The force sensors were placed in the environmental chamber and the % RH levels inside the chamber was programmed to vary from 20% RH to 80% RH, in steps of 20% RH at 23.5 °C. In addition, the temperature inside the chamber was increased from 55 °F to 73 °F to 85 °F for investigating the effect of temperature on the sensor. The sensor responses were sequentially recorded by an Instek 6100 LCR meter using Keithley 2700 mainframe with 7700 multiplexer. The multiplexer and LCR meter were connected to a desktop with MATLAB program via general-purpose interface bus (GPIB) and serial RS232 cables, respectively.

III. RESULTS AND DISCUSSION

Figure 4 shows the dynamic capacitive response of the sensor from 0 N to 10 N, in steps of 2 N and 20 N to 100 N, in steps of 20 N. Initially, no force was applied on the force sensor and its base capacitance value was recorded. Then, the force

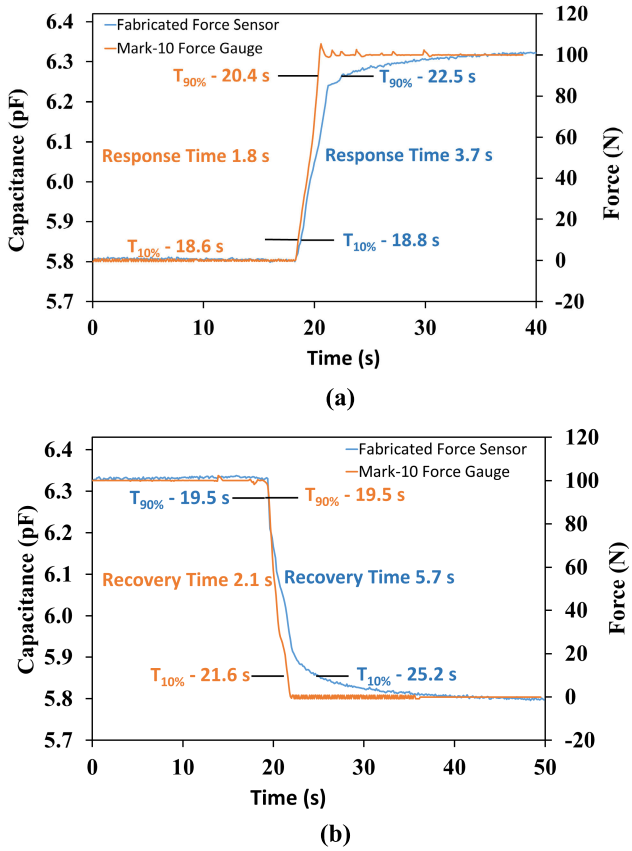

FIGURE 5. (a) Response time and (b) recovery time of the force sensor at full scale (0 to 100 N).

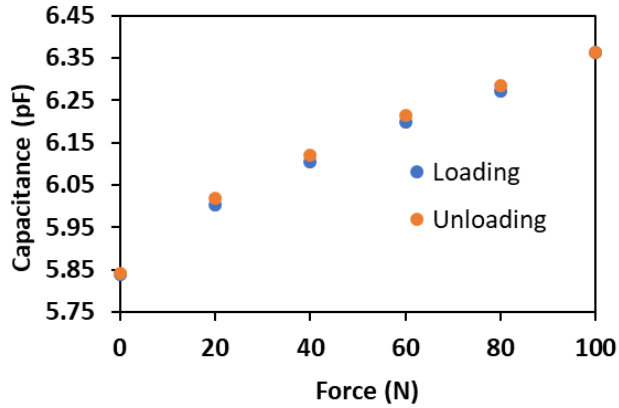

FIGURE 6. Hysteresis of the force sensor.

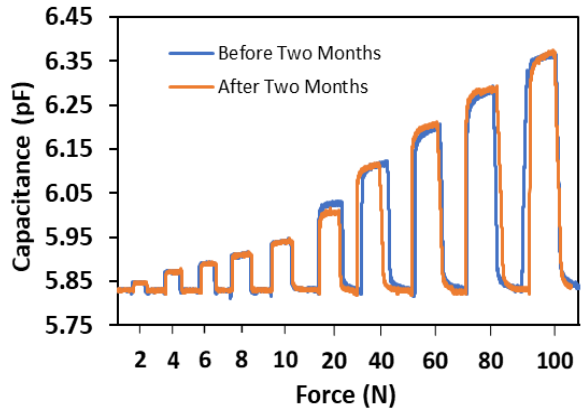

FIGURE 7. Stability results of the sensor for varying applied forces (0 to 100 N) before and after the two-month time duration.

sensor was subjected to the varying applied forces up to 100 N. It was observed that the capacitance increased from ~5.83 pF to ~5.85 pF to ~5.87 pF to ~5.89 pF to ~5.91 pF to ~5.94 pF to ~6.03 pF to ~6.11 pF to ~6.19 pF to ~6.27 pF to ~6.36 pF as the applied forces were increased from 0 N to 100 N, respectively (Fig. 4(a)). These results correspond to a relative capacitance change of ~0.3 %, ~0.7%, ~1.0 %, ~1.4 %, ~1.8 %, ~3.0 %, ~4.7%, ~6.4 %, ~7.7 %, ~9.5 % for 0 N to 100 N, when compared to the base capacitance value (Fig. 4(b)). A linear response with a sensitivity of $0.185\%N^{-1}$ and $0.081\%N^{-1}$ as well as a correlation coefficient of 0.998 were obtained for the force sensors subjected to applied forces 0 N to 10 N and 20 N to 100 N, respectively. It is worth noting that the fabricated force sensor attains its base value after removing the applied forces (six sensors; each three times tested).

The response time of the force sensor was measured by applying a force of 100 N using the Mark-10 force gauge (Fig. 5(a)). The capacitance of the sensor increased from 5.82 pF to 6.34 pF in 3.7 seconds. Similarly, the recovery time of the force sensor was measured when the applied force of 100 N was removed. This resulted in a recovery time of 5.7 seconds and the capacitance decreased from 6.34 pF to 5.81 pF (Fig. 5(b)). Then the hysteresis of the force sensors was investigated by loading and unloading the applied forces from 0 N to 100 N using Mark-10 force gauge (Fig. 6). The hysteresis of the sensor was mathematically calculated using Eq. (2) [47].

$$Max\ Hysteresis\ (\%fsd) = \frac{\max(O_{decreasingI} - O_{inreasingI})}{O_{max} - O_{min}} \tag{2}$$

where fsd - full-scale deflection, O and I is the output (y-axis) and input (x-axis) values of the sensor, respectively. A maximum hysteresis of 3.2% was calculated at the force of 40 N. This indicates that the force sensor has relatively better recovery and elasticity characteristics. Further, the stability of the sensor was investigated by storing the sensor samples (over 2 months) in a Plas-Labs 860-CGA acrylic vacuum cabinet with desiccants. Fig. 7 shows the stability results of the sensor for varying applied pressures (0 to 100 N) before and after the two-month time duration. A maximum change of 0.5% was calculated at 20 N, before and after the two-month time period, indicating that the fabricated force sensor is very stable for over two months.

In addition to hysteresis and stability tests, the effect of humidity on the sensor performance was investigated by varying % RH levels from 20% RH to 80% RH, in steps of 20% RH at 23.5 °C in a programmable Thermotron® SE-3000 environmental chamber (Fig. 8). It was observed

TABLE 1. Summary of recently published research work on force sensors.

Material	Fabrication Method	Range	Size	Sensitivity	Response Time	Recovery Time	Hysteresis/ Stability	Ref.
Graphite/PDMS	3D Printing/Casting	3.5 -17.5 mN	30 × 30 mm	0.2542 pF/mN	-	-	-	[41]
Al/PDMS/PMMA	Laser Ablation/Casting	0.6 – 7.2 N	6 × 5 cm	1.03 V/N	-	-	-	[42]
Paper/MoS_2/Graphite	Doctor blade (or tape casting)	2.45 - 9.8 N	9 cm^2	0.18 V/N	10 ms	-	No hysteresis studies/Stable over 30 days	[43]
PET/ITO-PDMS	Spin Coating/Casting	1 – 10 N	10 × 10 mm	0.011 N^{-1}	200 ms	-	Max. hysteresis – 6.49%	[44]
CNT/PDMS	Screen printing/Casting	0 - 41 N	30 × 10 mm	0.029%/N	-	-	-	[35]
EGaIn/PDMS	Casting	0 - 198 N	20 × 9 mm	0.00061%/N	-	-	Stable for 2 months	[36]
Graphene/Silver nanowires	Dip coating	0 - 16 N	2 × 2 cm	0.04/N	54 ms	-	No hysteresis studies/Stable	[51]
Constantan	MEMS	20 – 100 kN	1.2 cm^2	0.0317 V/V	N/A	-	Hysteresis – 0.047%	[52]
Ag/PDMS	Screen printing/Casting	0-10 N 0-100 N	Ø15 mm	0.185%/N 0.081%/N	600 ms 3.7 s	1.2 s 5.7 s	Max. hysteresis – 3.2%/ Stability tests performed for 2 months – 0.5%	This work

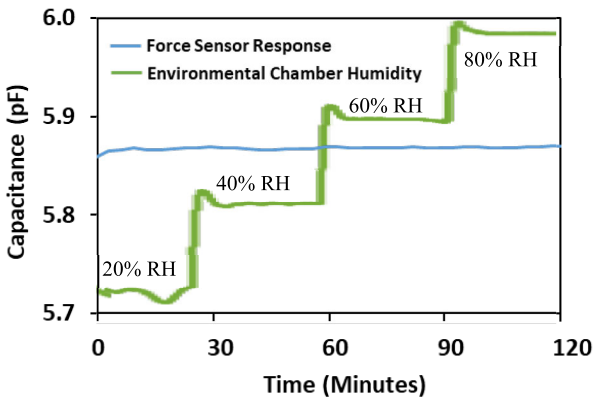

FIGURE 8. Effect of relative humidity varying from 20% RH to 80% RH on the force sensor.

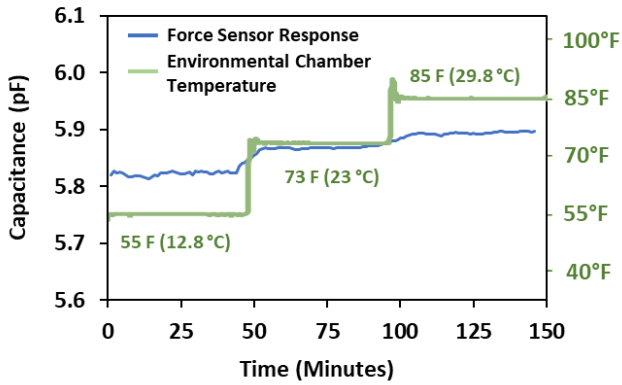

FIGURE 9. Effect of temperature on the force sensor.

that the base capacitance of the force sensor increased from 5.83 pF to 5.86 pF due to accumulation of parasitic or stray capacitances in the connecting cables. The varying humidity levels had minimal effect on the sensor response. Apart from humidity, the effect of temperature on the sensor was also investigated by varying the temperature of the environmental chamber from 55 °F (12.8 °C) to 73 °F (23 °C) to 85 °F (29.8 °C). It was observed that the capacitance increased from 5.81 pF to 5.84 pF to 5.86 pF when the temperature of the environmental chamber was increased from 55 °F to 73 °F to 85 °F, respectively and affects the capacitance measurements of forces below 10 N (Fig. 9). This effect can be attributed to the thermal expansion coefficient of the PDMS dielectric layer that leads to micro-thermal deformations in the sensor [48], [49]. The temperature effect on the force sensor response can be mitigated using well known reference gauge compensation techniques [49], [50]. In this technique, a reference force sensor (same as the printed force sensor) placed under the same ambient conditions of force sensor serves as a reference capacitor. The reference force sensor is not subjected to external applied forces. During the capacitance measurements, the absolute capacitance value of the reference force sensor is deducted from the printed force sensor at any given time (under both loading and unloading of forces) to alleviate/mitigate the effect of temperature.

A comparison table of some recently reported force sensors along with the current work is provided in Table 1. It is worth noting that the printed force sensor has better sensitivity and stability when compared to some recently reported force sensors. The results thus demonstrated the feasibility of employing a polyimide based flexible capacitive force sensor that was fabricated using additive screen-printing process and 16:1 PDMS as an efficient and cost-effective way to monitor forces in wide variety of applications.

IV. CONCLUSION

In this work, a capacitive force sensor was successfully fabricated on a flexible polyimide platform. The top and bottom electrodes of the sensor were fabricated by screen printing Ag ink on polyimide substrate. A PDMS based dielectric layer was prepared by mixing pre-polymer and curing agent in 16:1 (w/w) ratio. The force sensor was assembled by attaching the top and bottom electrodes to the PDMS dielectric layer using a double-sided polyimide tape. An increase in capacitance from 5.83 pF to 6.36 pF was observed when the applied forces were increased from 0 N to 100 N resulting in an overall relative capacitance change of 9.5%. A sensitivity of 0.081%/N^{-1}, a response time of 3.7s and a recovery time of 5.7s was obtained for the force sensor. Future work includes implementation of reference gauge compensation technique to mitigate the effect of temperature on sensor as well as investigate the mechanical stresses such as bending on the performance of the force sensor.

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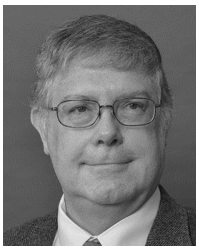
DINESH MADDIPATLA (Student Member, IEEE) received the B.E. degree in electrical and electronics engineering from Anna University, India, in 2013, and the M.Sc. degree in electrical engineering and the Ph.D. degree in electrical and computer engineering from Western Michigan University, Kalamazoo, USA, in 2016 and 2020, respectively. He is currently working as a Research Associate with the Centre for Advanced Smart Sensors and Structures (CASSS), Western Michigan University. He has published over 70 refereed journal articles and refereed conference proceedings. In addition, he has published over 15 patents and invention disclosures. His research interests include all aspects of design, fabrication, and characterization of printed electronics focusing on flexible sensor structures, microfluidics, electrochemical sensors, gas sensors, and lab-on-a-chip sensing systems. He has been serving as a Reviewer for more than 15 international journals. He was awarded as the Technology Transfer Talent Network (T3N) from the State of Michigan, the All-University Research and Creative Scholar Award, and the Dissertation Completion Fellowship from Western Michigan University.

XINGZHE ZHANG received the B.S. degree in electronics engineering from Mapua University, Manila, Philippines, in 2014, and the M.Sc. degree in electrical engineering from Western Michigan University, Kalamazoo, USA, in 2017, where he is currently pursuing the Ph.D. degree in electrical engineering. His research interests include design, fabrication, and characterization of printed sensor, printed electronics, biomedical sensing, lab-on-a-chip, and smart sensing.

ARNESH K. BOSE (Graduate Student Member, IEEE) received the B.Tech. degree from the West Bengal University of Technology, Kolkata, India, and the M.S. degree from the University of Kansas, Lawrence, KS, USA, in 2016. He is currently pursuing the Ph.D. degree in electrical engineering with Western Michigan University, Kalamazoo, MI, USA. He is currently a Teaching Assistant with the Department of Electrical and Computer Engineering, College of Engineering and Applied Sciences, Western Michigan University. He is also a Research Assistant with the Center for Advanced Smart Sensors and Structures, Department of Electrical and Computer Engineering. His current research interests include design, fabrication, and charaterization of printed sensors, laser-assisted patterning of thin films, flexible microplsma discharges devices, surface sterilization of biofilms, plasma-based liquid media sterilization, and optical-based electrochemical sensing.

SIMIN MASIHI (Graduate Student Member, IEEE) received the B.S. degree from Zanjan University, Zanjan, Iran, in 2010, and the M.S. degree from the Adiban Institute of Higher Education, Garmsar, Iran, in 2016, all in electrical engineering. She is currently pursuing the Ph.D. degree in electronics engineering with Western Michigan University, Kalamazoo, MI, USA. She is currently working as a Research Assistant with the Center for Advanced Smart Sensors and Structures, Department of Electrical and Computer Engineering. Her current research interests include design, fabrication, and development of printable wearable microstrip antennas, biomedical applications, sensors, development of dielectric materials, laser-assisted patterning of thin films, and flexible hybrid electronics.

BINU B. NARAKATHU (Member, IEEE) received the B.E. degree in electronics and communication engineering from Visvesvaraya Technological University, Bengaluru, India, the M.Sc. degree in computer engineering from Western Michigan University, USA, in 2009, and the Ph.D. degree from the Department of Electrical and Computer Engineering, Western Michigan University, in 2014. From 2015 to 2017, he was a Postdoctoral Fellow with the Sensor Technology Laboratory (STL), Department of Electrical and Computer Engineering, Western Michigan University. He is currently a Research Associate with the Center for Advanced Smart Sensors and Structures (CASSS), Department of Electrical and Computer Engineering, Western Michigan University. His research interests include all aspects of design, fabrication, and characterization of high-performance sensing systems, microfluidic devices, lab-on-a-chip for point-of-care testing (POCT), biosensors, bioelectronics, printed electronic devices, and BioMEMS devices for applications in the biomedical, environmental, and defense industries. He has published over 140 refereed journal articles and refereed conference proceedings. In addition, he has published ten patents and more than 30 invention disclosures. He is serving as a member of the Technical Council Committee for NextFlex, a DoD funded consortium for Flexible Hybrid Electronics.



BRADLEY J. BAZUIN (Member, IEEE) received the B.S. degree in electrical engineering from Yale University, New Haven, CT, USA, in 1980, and the M.S. and Ph.D. degrees in electrical engineering from Stanford University, Stanford, CA, USA, in 1982 and 1989, respectively. He was a Research Assistant with the Center for Integrated Electronics in Medicine associated with the Integrates Circuits Laboratory, Center for Integrated Systems, Stanford University, from 1981 to 1988, a part-time MTS and System Engineer, from 1981 to 1989, a Principal Engineer with ARGOSystems, Sunnyvale, CA, USA, from 1989 to 1991, and a Senior Systems Engineer with Radix Technologies, Mountain View, CA, USA, from 1991 to 2000. Since 2000, he has been a Term Appointed Assistant Professor and an Assistant Professor with Western Michigan University, Kalamazoo, MI, USA, where he is currently an Associate Professor of electrical and computer engineering. His current research interests include printed electronics, electronic and printed circuit board circuit design and fabrication, custom integrated circuit design, embedded signal processing, wireless communication, software defined radios, and advanced digital signal processing algorithms for physical layer communication systems. He is a member of the American Society for Engineering Education and the Institute of Navigation.

JOHN D. WILLIAMS is currently the Lead Engineer for additive electronics manufacturing at Boeing Research and Technology (BR&T), Huntsville. After completing his Ph.D. degree, he has served as the Science and Technology Lead for the Metal Micromachining effort at Sandia National Laboratories, Albuquerque, NM. He later joined The University of Alabama, Huntsville, as an Assistant Professor of electrical, materials, and optical engineering, and the Associate Director of the Nano and Micro Devices Center. He has more than 15 years of experience as the PI Lead for prototype development of microelectromechanical systems (MEMS) with over 13 patents and 40 peer reviewed publications in the areas of materials processing, metals, microfabrication, photonics, and RF electronics and bioMEMS. His research has been funded by NSF, DARPA, NASA, NIH, and U.S. ARMY-SMDC. He is currently the Principal Investigator on three NextFlex projects. One to develop flexible antenna arrays, a second to improve off axis printing using an nScrypt printer, and a third to develop the MRL for passive electronic devices. He joined Boeing' Research and Technology Division, in June 2014, to support new research efforts for RF design and fabrication.

MICHAEL F. MITCHELL received the Ph.D. degree from the University of Florida, in 2015, and specializes in analog electronics, data acquisition, and signal processing. He is currently an Electrical Engineer with the Additive Electronics Technologies (AET) Group, Boeing Research and Technology (BR&T), Huntsville, AL. In 2018, he joined BR&T's AET Group and applies his specialization to flexible hybrid electronics and is the PI for NextFlex Project Call 3.5, Condition Monitoring Sensor Array.

MASSOOD Z. ATASHBAR (Senior Member, IEEE) received the B.Sc. degree in electrical engineering from the Isfahan University of Technology, Isfahan, Iran, the M.Sc. degree in electrical engineering from the Sharif University of Technology, Tehran, Iran, and the Ph.D. degree from the Department of Communication and Electronic Engineering, Royal Melbourne Institute of Technology University, Melbourne, Australia, in 1998. From 1998 to 1999, he was a Postdoctoral Fellow with the Center for Electronic Engineering and Acoustic Materials, The Pennsylvania State University, University Park. He is currently a Professor with the Electrical and Computer Engineering Department and the Founding Director of the Center for Advanced Smart Sensors and Structures (CASSS), Western Michigan University, USA. His research interests include physical and chemical sensor development, wireless sensors, and applications of nanotechnology in sensors, and flexible hybrid electronic devices. He has published over 270 refereed articles, refereed conference proceedings, and four book chapters. In addition, he has published ten patents and more than 30 invention disclosures. He is serving as a Lead Member of the Technical Council Committee and is the Co-Chair for the Technical Working Group (TWG), NextFlex, a DoD funded consortium for Flexible Hybrid Electronics. He was a member of the Editorial Board and an Associate Editor for the journal of IEEE Sensors, from 2006 to 2016.